Event-Driven Sentiment Analysis of Indian Audiences Towards Minority Content Creators

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Research Question

Primary research question-
How do Indian audiences' sentiments towards Muslim content creators change over time following major geopolitical events?

Sub-questions-

- Do geopolitical events (like the Pathankot attack, the Uri strike, the Pulwama attack, the Pahalgam attack and Operation Sindoor) trigger significant shifts in Indian audiences' sentiments even in the non-political sphere?
- Is the perspective positive, critical, or neutral? How much does the volume of comments change around these periods?
- What themes dominate these discussions (such as nationalism, hostility, support)?

Data Sources

Primarily our data will come from comments sections under content posted by Indian Muslim content creators. We will select 5-10 Muslim content creators whose content is purely for entertainment purposes. Some of the data sources we will be looking at are:

- YouTube comments on Indian Muslim content creators' channels
- Reddit discussions following these geopolitical events
- X (Twitter) posts and replies using relevant hashtags
- Instagram comments under content posted by these creators

Feasibility of the Data

The data will be collected using data from YouTube comments APIs, X, Reddit etc. where comments threads, timestamps and tweets will be accessed.
The dataset is restricted to a particular number of creators from the community of interest and a particular number of videos. Comments will also be sampled strategically.

Scalability-

The dataset will be made using approximately 10k-50k comments with the help of API, given the constraints involved in accessing them we hope that this will be sufficient to draw conclusions.

Large Language Models and Natural Language Processing Models-
Given the diversity in the Indian crowd, pretrained multilingual models will be used which can handle mixed languages easily, such as Hinglish (Hindi written in English format). Tasks like sentiment analysis and topic detections can be performed using regular GPU or cloud based platforms (such as Google Colab). Additionally, LLM APIs such as chat GPT, Google Gemini or open-source models may be used to support analyses like contextual identification.

Methodology & Analysis

- The data will be cleaned to remove noise and other non-text data like emojis. Multilingual content will be detected and handled accordingly.
- For the sentiment analysis, comments will be classified into 3 categories, positive, negative and neutral. Additionally, comments using toxic and hostile language will be flagged.
- Comments will be classified under themes like nationalism, religion and political commentary.
- Since this is a temporal analysis of the change in public sentiments around the occurrence of a major geopolitical event, closer attention will be paid to pre-event and post-event sentiments to identify any spikes or anomalies in the data trend.

Expected Results

Following the analyses, we expect to see:

- A shift in the tone of comments under content creator videos from apolitical to more political and religious topics around the periods of geopolitical events.
- Increased polarisation of audiences around such events, with a heavy concentration of negative comments.
- Occurrence of repeated themes in the discussions such as nationalism, identity, and hostile or supportive expressions.

To supplement our findings, we will produce:

- Time-series analysis showing fluctuations in sentiments aligned with the geopolitical news.
- Distributional analysis of positive, negative and neutral comments.